

Annealing for the improvement of the capabilities of parylene C as electret

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ABSTRACT: Parylene C is a polymer elaborated by chemical vapor deposition and is particularly useful for the coating of deformed structures. New-generation soft energy harvesting electrostatic generators can be of complex shape. These generators need electret material in order for them to function. It could thus be interesting to incorporate parylene C in this device. However, this polymer suffers from bad retention of trapped charges after corona discharge and a poor thermal stability of these charges. In this work, parylene C polymers were subjected to specific annealing before corona charging. This induces strong changes in the crystallinity of the material. Atomic force microscopy and X-ray diffraction quantified the crystalline nature of parylene C following the annealing temperature. Surface potential decay of corona-charged parylene C demonstrated that annealing to about 150°C is beneficial for the trapping of charges for a long time (analysis over 12 days). At the same time, thermal stability of the electret parylene C is achieved up to 100°C. Spherulites with a specific area close to the surface of the polymer could explain such performances due to their influence on the distribution of traps. Dielectric analysis does not confirm that the decay of surface charges is correlated to the dynamics of polymer chains, although the moving of chains probably influences the flowing of these charges. In sum, annealed parylene C appears to be a new interesting candidate for electret-based electrostatic generators. © 2018 Wiley Periodicals, Inc. *J. Appl. Polym. Sci.* **2018**, 135, 46908.

KEYWORDS: corona discharge; electret; parylene C; semi-crystalline polymer; surface potential decay; thermal stability

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INTRODUCTION

Electrets are dielectric insulating materials, generally polymers, with quasi-permanent electrical charges.^{1–4} They are used in many applications requiring a permanent electric field without any permanent voltage source.^{5–7} Promising applications using electrets include electrostatic generators for integration in micro-electro-mechanical systems^{8,9} and for large-scale devices.^{10,11} Halogenated polymers such as Teflon, poly(vinylidene fluoride) and its copolymers have emerged as essential electret materials^{12–14} and for integration in the energy harvesting field through hybrid electrostatic generators. In recent decades, parylene C has been regarded as a potential electret candidate for energy harvesting.^{15,16} Today, the use of parylene C is essential for many industrial applications due to the well-controlled chemical vapor deposition (CVD) process compatible with microelectronic applications, its excellent conformal deposition in very complex geometries, its excellent (di)electric properties (low conductivity, low dielectric losses and low leakage current) and its barrier properties against various chemical agents. Furthermore, it is biocompatible and its deposit remains uniform even over large areas.^{17,18} Unfortunately, this polymer is strongly polar due

to the C–Cl bonds present in its chemical structure, and the main problem with parylene C electrets is the stability of electric charges over time. Several methods have been proposed for improving the electret stability, including chemical modification of the polymer surface^{19,20} and surface plasma treatment at low temperature.²¹ Another approach consists in using porous or textured materials intended to increase the density of charges implanted in the material.^{22,23} Although the charge stability is clearly improved with these methods, unfortunately two key parameters contributing to the capabilities of electrets are affected: the amount of charge retained and the thermal stability. A different approach, which is easily applicable, consists in annealing the material before the electric charging process.^{24,25} The measurement of surface potential (SP) and surface potential decay (SPD) for a linear temperature ramp was used to estimate the electret surface charge density and thermal stability.^{26,27}

Herein, we improved applications of parylene C polymers as electrets. These later are fabricated after depositing ions at room temperature by corona discharge on their surface. Annealing performed on the material before depositing charges induces changes in the crystallinity, in the surface roughness and in the

dipolar nature of molecular chains. Electret performance is strongly influenced by annealing: specific thermal conditions improve both the thermal stability and the charge density of these parylene-based electrets over time. This is analyzed according to the physico-chemical, morphological and dielectric changes occurring in the bulk and on the surface of the annealed material. Finally, a thermally stimulated current allows these capabilities to be interpreted from shallow and deep traps in parylene C.

EXPERIMENTAL

Materials and Annealing Electret Preparation

Parylene C is deposited at room temperature on aluminum substrates by CVD. It is obtained from the dichloro-di-*p*-xylylene dimer using the Gorham process.²⁸ The deposition process is carried out in three steps. The first step is the sublimation, which converts the solid dimer into the gaseous phase at a temperature of between 120 and 140 °C and a pressure of 0.5 mbar. Then the gaseous molecules pass into the pyrolysis chamber at a temperature of 690 °C and a pressure of 0.1 mbar. In the third step, all the monomers (Figure 1) from the pyrolysis chamber at a pressure of 0.02 mbar polymerize on the substrate. The thickness of the films, measured in situ in the chamber, is 4.5 μm.

For the parylene C annealing process, a temperature controller (Linkam TMS94) was used. The annealing temperature T_a is reached with a ramp of 10 °C/min from room temperature. Samples were kept at T_a for 6 h before being cooled down to room temperature with a ramp of −1 °C/min. Table I lists the materials with different annealing temperatures (the annealing process was realized in different samples deposited in the same conditions). The justification of the annealing temperatures will be presented in the results section.

After the annealing-cooling process, some samples were charged electrically at room temperature by corona discharge (Figure 2).

A potential of −1000 V was applied to the needle for 180 s, thereby generating ions accelerated towards the grounded sample. The negative charges implanted in the electret establish a SP V_s . The grid voltage V_g allows the charges deposited on the surface to be fixed and distributed. The electrical charging is carried out on samples with a surface of $3 \times 3 \text{ cm}^2$.

The electret SP was measured with an electrostatic voltmeter (Model 244, Monroe Electronics) with the manufacturer-recommended distance of 3 mm between the measurement needle and the sample. After being subjected to a corona discharge, the parylene C will be referred to as electret parylene C in the rest of this work.

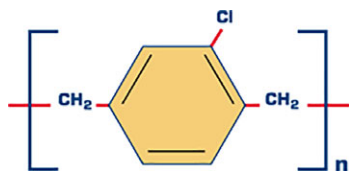


Figure 1. Chemical structure of the parylene C monomer. [Color figure can be viewed at wileyonlinelibrary.com]

Table I. Annealed Parylene C Materials

Parylene C (4.5 μm)	Annealing temperature (°C)
A	Not annealed (as deposited)
B	60
C	150
D	250
E	350

Chemical and Morphological Analysis

Differential scanning calorimetry (DSC) using DSC TA Q20 equipment was performed for the as-deposited samples to identify the phase transition temperatures and also to define different annealing temperatures contributing to the electret stability enhancement. Thermogravimetric analysis (TGA) is carried out to identify the degradation temperature of parylene C under nitrogen gas.

The chemical composition of parylene C before and after annealing was monitored by Fourier transform infrared (FTIR) spectroscopy in the wavenumber range 400–1800 cm^{-1} using a Nicolet 380 spectrometer. The bulk crystallinity is measured by X-ray diffraction using a high-resolution (θ – θ) Brükker D8 Advance diffractometer equipped with an X-ray tube with copper anticathode $\text{CuK}\alpha$ ($\lambda = 1.540562 \text{ \AA}$, 40 kV, 30 mA). The diffraction diagram is obtained from 5° to 30° in 2θ with a 0.02° step and a 2 s counting time. The morphology is provided by atomic force microscopy (Dimension Icon AFM, Bruker). Average surface spherulite areas and surface roughness will be extracted from these measurements.

Dielectric Analysis

For energy harvesting electrostatic generators based on the time variation of the capacity $C(t)$ of the system, the harvested power P depends on the dielectric properties according to eq. (1)²⁹.

$$P \propto \left(\frac{\rho d}{\epsilon_0 \epsilon_r} \right)^2 \frac{dc}{dt} \quad (1)$$

where ρ is the surface charge density of the electret, d the thickness and ϵ_r the relative permittivity. ϵ_0 the vacuum permittivity ($8.854 \times 10^{-12} \text{ F/m}$). As-deposited and annealed parylene C samples are in a metal–insulator–metal (MIM) configuration through the deposition of top gold electrodes and polarized with an alternating voltage of 1 V_{rms} for a frequency ranging from 10^{-1} to 10^6 Hz using a Novocontrol Alpha impedance analyzer.

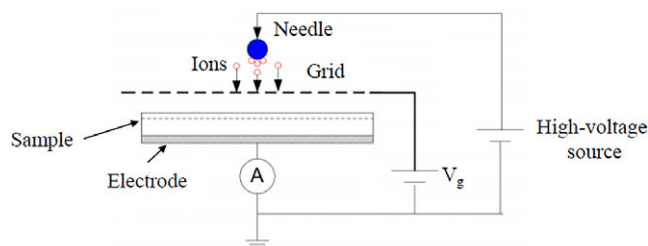


Figure 2. Corona discharge experimental set-up. [Color figure can be viewed at wileyonlinelibrary.com]

Measurements are taken from a surrounding wall working under a primary vacuum chamber and under nitrogen gas. No corona discharge was previously reached on the parylene C for this dielectric analysis.

RESULTS AND DISCUSSION

The first part of this study consists in the choice of representative and judicious annealing temperatures. For that, DSC was done on the as-deposited parylene C (Figure 3). It reveals three typical temperatures: the first one is the glass transition temperature $T_g = 92^\circ\text{C}$, the second one is the crystallization temperature $T_c = 155^\circ\text{C}$ and the last one corresponds to the melting temperature $T_m = 301^\circ\text{C}$, all of which are in line with the results obtained in previous works.³⁰ Based on this study, annealing below T_g was carried out at 60°C (sample B), annealing at 150°C between T_g and T_c (sample C), annealing at 250°C between T_c and T_m (sample D) and finally an annealing at 350°C above T_m (sample E). Sample A is the reference sample (un-annealed).

The TGA thermogram [Figure 3(b)] shows that the as-deposited parylene C is stable up to 400°C as already observed in previous work,³¹ suggesting that parylene C can be used up to this high temperature. This stability is however not so obvious, as further demonstrated by FTIR analysis.

The main aim of this study is to investigate the influence of annealing on the capabilities of parylene C as an electret. Thus, SP measurements (Figure 4) were carried out over 12 days for each sample (using an electrostatic voltmeter) in order to reach the potential stability and get rid of surplus electrical charges.

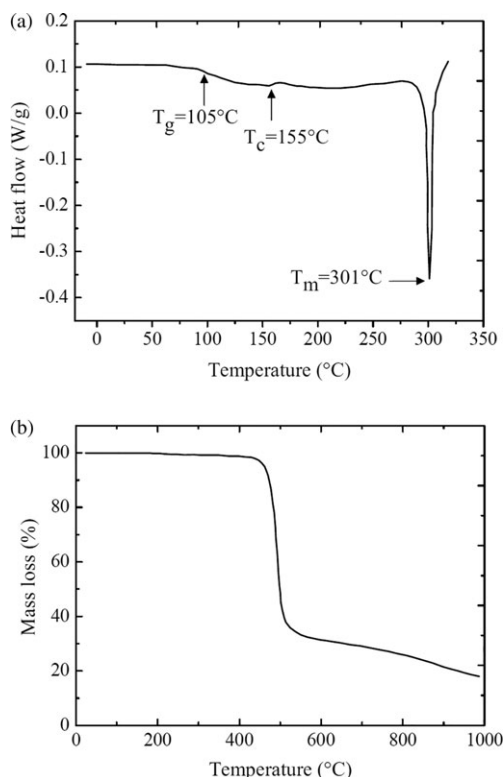


Figure 3. (a) DSC and (b) TGA thermograms of the as-deposited parylene C (sample A) (a low mass loss of order of 1% at 420°C).

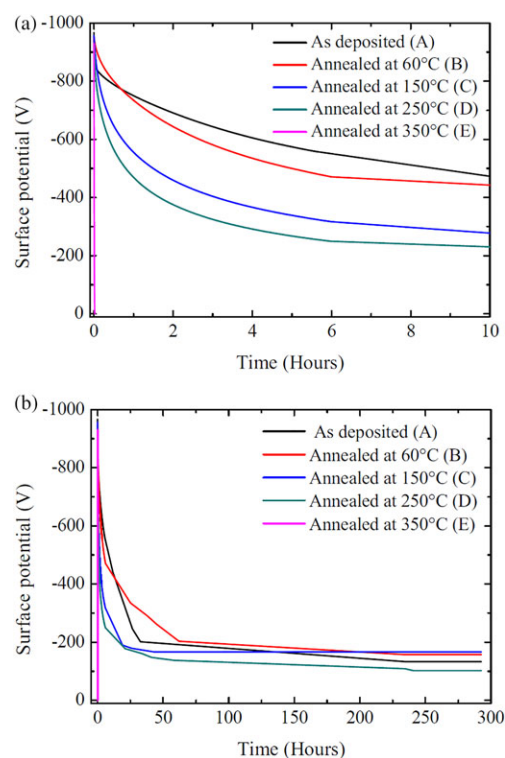


Figure 4. Surface potential decay electret parylene C (a) at short times and (b) at longer times after charging [(a) is a zoom of (b) for the short times]. [Color figure can be viewed at wileyonlinelibrary.com]

These latter are deposited in the traps, especially at the sample surface, as was highlighted by Gang-Jin *et al.*³²

Sample E with atypical instantaneous decay will be discussed later. Other samples present a strong decrease during the first 2 days following the charging process. At longer times, this decay is reduced and it is close to stabilization after 6 days. Values between $|100|$ and $|250|$ V are obtained following the annealing, thereby corresponding to a reduction of $75\% \pm 5\%$ of the initial potential value V_0 . This final value may appear low, but it is high enough for integration in electrostatic generators.^{15,33} For example, Chiu and Wu³⁴ developed an electrostatic energy harvester based on parylene C electret with a SP of 200 V³⁴. To increase more the SP, stronger corona discharge voltage could be applied or the manufacture of thicker electrets.

In order to analyze the origin of this decay and its correlation with changes induced in parylene C after annealing, chemical structure identification using FTIR was performed before and after the annealing process (Figure 5). For the samples annealed at 60 , 150 , and 250°C (samples B, C, and D, respectively), parylene C remains stable without any chemical degradation. The FTIR spectrum of the annealed sample at 350°C (sample E) shows five missing bands: bands at 1107 , 1155 , and 1213 cm^{-1} are assigned to the aliphatic C—H bonds, and bands at 448 and 603 cm^{-1} are assigned to the C—C aliphatic bonds compared to the as-deposited parylene C. These massive changes indicate a degradation in the chemical structure of this material annealed above melting point, as was also recently identified by Von Metzzen *et al.*³⁵ This was not surprising as it exhibited a brown-

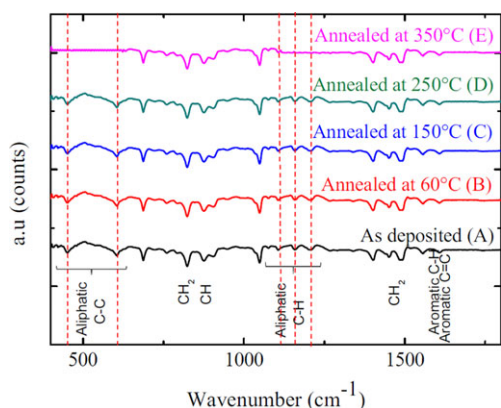


Figure 5. FTIR spectra of as-deposited and annealed parylene C samples. [Color figure can be viewed at wileyonlinelibrary.com]

golden shiny surface and could not be peeled off from the substrate. This explains why the SP decay in this sample was abrupt (Figure 4, sample E) and it will not be considered for future discussions in this article.

It is well known that annealing of semi-crystalline polymers strongly affects electric charge conduction on the surface and in the bulk. In order to evaluate the impact of semi-crystalline changes on the SP decay, XRD measurements were performed on virgin and annealed parylene C samples. This material is a semi-crystalline polymer with a typical bulk crystallinity of 45% as confirmed by XRD in our as-deposited parylene C. No change in the crystalline orientation was observed after the annealing-cooling process (Figure 6). In addition, peaks shift to larger angles with the increase in the annealing temperature. This trend is representative of a modification in the distribution of the crystalline lamellae and thus a change in the percentage of the crystalline phase and the size of the lamellae. The Bragg relation expressed by eq. (2) allows the inter-chain distance d to be determined.³⁶

$$2d \sin \theta = q\lambda \quad (2)$$

where q is the diffraction order, λ is the wavelength, and θ is the diffraction angle.

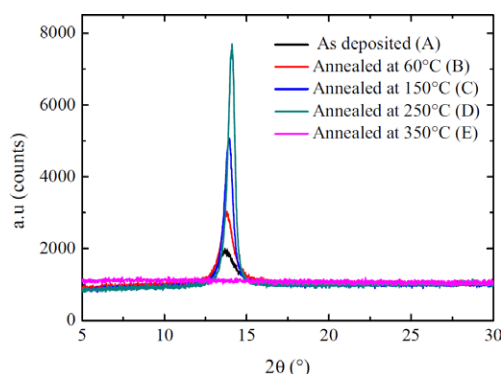


Figure 6. XRD pattern of the parylene C (impact of the post-annealing process). [Color figure can be viewed at wileyonlinelibrary.com]

The Debye–Scherrer formula given by eq. (3) was used to calculate the average size of the crystallites.³⁷

$$\delta = \frac{K\lambda}{\text{FWHM} \cos \theta} \quad (3)$$

where K = constant = 0.9, FWHM is the width at half height of the peaks and δ is the grain size.

Figure 7 reports the bulk crystallinity rate χ_c , the average size of crystallites δ and the inter-chain distance d as a function of the annealing temperature. As expected after annealing of semi-crystalline polymers below the melting temperature, one observes at the same time an increase in the crystallinity rate and crystallite size and a reduction in the inter-chain distance at higher annealing temperature. The same behavior was reported by Kim *et al.*³⁸ Very slight differences are found in the reported values and this can be attributed to the difference in the used substrates and the conditions (temperature and pressure) in the CVD process of the development of the materials. Similar behavior has been reported for parylene HT by Diaham *et al.*³⁹ AFM experiments confirm the influence of annealing on the semi-crystalline surface morphology and spherulites are clearly identified (Figure 8).

Image processing of AFM maps allows the mean area of the spherulites and the average surface roughness of parylene C to be estimated (Figure 9). Both increase with the annealing temperature to reach a maximum between 150 and 250°C near the crystallization temperature during the heating cycle.

It is known that for this type of semi-crystalline polymer that close to the crystallization temperature of the bulk material, crystallization and rapid progressive germination occur at the surface of the material.⁴⁰ This phenomenon occurs in a temperature range not exceeding 15°C before the bulk crystallization and it is exacerbated in the case of relatively thick films that exceed a few 100 nm. Indeed, thick films prevent the influence of the substrate, which restricts the molecular mobility in the bulk layer of the polymer at the bottom interface.⁴⁰ This will contribute to the increase of the average surface roughness as can be seen in Figure 9. In addition, when the annealing temperature exceeds the crystallization temperature (155°C), surface crystallites will be progressively broken down, thereby inducing both smaller spherulite areas and average surface roughness.

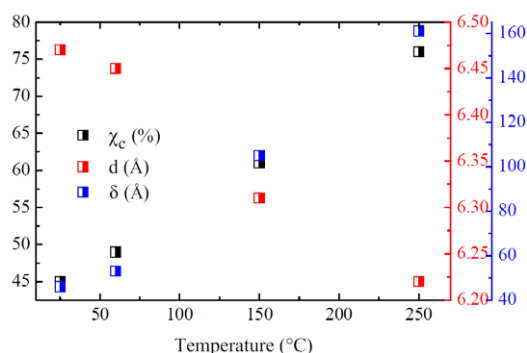


Figure 7. Crystallinity [χ_c (%)], size of crystallites [δ (Å)] and inter-chain distance [d (Å)] as a function of the annealing temperature. [Color figure can be viewed at wileyonlinelibrary.com]

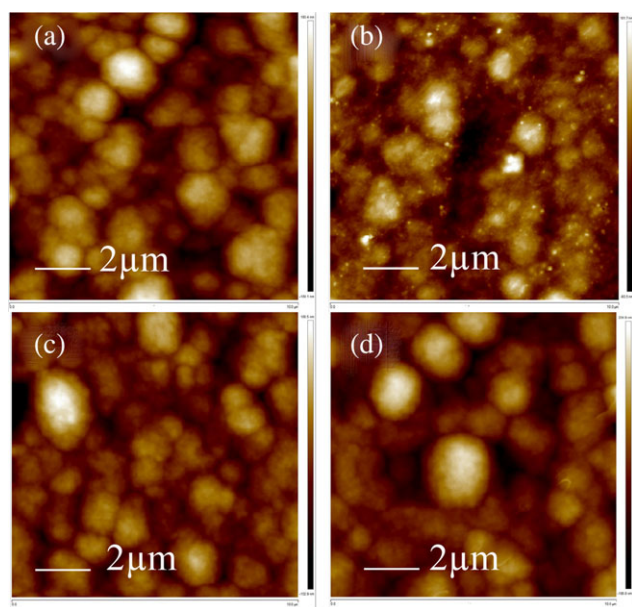


Figure 8. AFM topography $10 \times 10 \mu\text{m}$ of parylene C for different annealing temperatures [(a) the as-deposited sample, (b) the annealed sample at 60°C , (c) the annealed sample at 150°C , and (d) the annealed sample at 250°C]. [Color figure can be viewed at wileyonlinelibrary.com]

The relationship between the SPD and semi-crystalline behavior of parylene C is here considered again, focusing first on SPD during the first few days [Figure 4(a)]. One can assume that the surface state of parylene C influences the charge decay. As regards average surface roughness, samples A and D present an equivalent state whereas they show a great difference in SPD behavior. With regard to the spherulite area, the same conclusion can be conducted: sample C presents the biggest spherulite surface but sample D presents the highest SPD, whereas samples A and B have the lowest SPD with the smallest spherulite area. Thus, it is not clear whether a possible correlation exists between the surface state of parylene C and the SPD at first hours.

Yet, according to De Cupere *et al.*,⁴¹ Risnes *et al.*,⁴² De Rovère *et al.*,⁴³ Wang *et al.*,⁴⁴ and Suzuki *et al.*,²³ the electret SP and charge retention depend on the surface state. In fact, the SP increases with the average surface roughness and the spherulite area, thereby favoring the charge carriers' retention. This

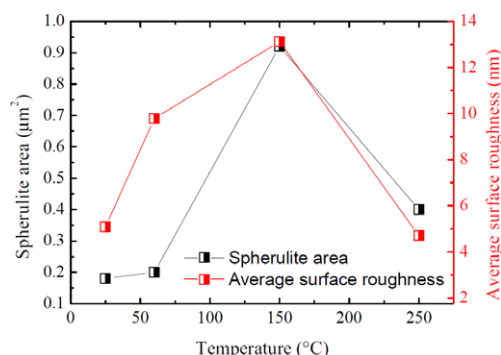


Figure 9. Average area of spherulites and average surface roughness before and after annealing. [Color figure can be viewed at wileyonlinelibrary.com]

phenomenon was explained by Shinotsuka *et al.*⁴⁰ in relation to semi-crystalline PET and PEN using the AFM technique through the increase of the surface crystallinity close to the crystallization temperature T_c . This could explain why sample C annealed at 150°C presents the highest SP after reaching the stable limit value [Figure 4(b)] but not the SPD at short times [Figure 4(a)].

Another possibility for explaining the surface decay in polar polymers relates to dipolar charges inside the material. Indeed, the deposit of surface charges by corona discharge induces an electric field through the material, thereby favoring the rotation of dipoles. The rotation of dipoles compensates for surface charges and consequently SP will decrease. To evaluate this assumption, dielectric spectroscopy measurements were conducted on virgin and annealed parylene C materials. The results are reported in terms of dielectric constant ϵ' and loss factor ϵ'' (Figure 10).

A dispersion of ϵ' as a function of frequency and a spreading peak of ϵ'' are clearly observed (especially clear for sample A, which exhibits the lowest crystallinity). The frequency dependence of ϵ'' for parylene C is mainly due to the C–Cl polar bonds of the chemical structure of parylene C (Figure 1). The relaxation peak associated with this dispersion of ϵ'' is attributed to secondary intermolecular relaxation in the amorphous part of the material (β -relaxation).^{45,46} We can see clearly that the dielectric constant ϵ' decreases with the annealing temperature. At the same time, the frequency relative to the peak of the losses shifts

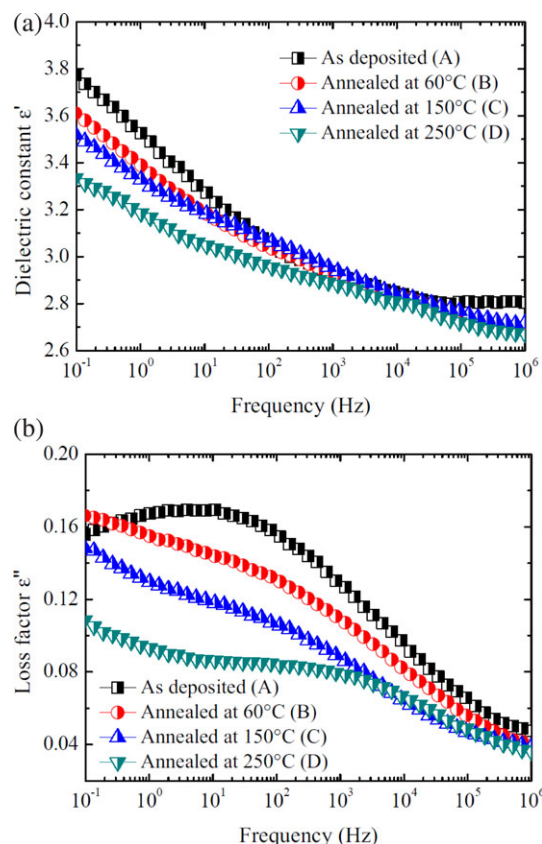


Figure 10. Dielectric constant ϵ' and loss factor ϵ'' versus frequency at room temperature for virgin and annealed parylene C. [Color figure can be viewed at wileyonlinelibrary.com]

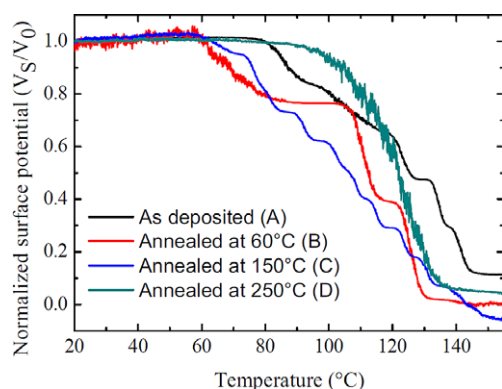


Figure 11. Thermally stimulated surface potential decay (TSSPD) for electret parylene C. [Color figure can be viewed at wileyonlinelibrary.com]

towards the high frequencies with a decrease in its amplitude. This trend in the dielectric response is usual for semi-crystalline polymers with highest crystallinity rate and crystallite size (see Figure 7): in this configuration, fewer amorphous chains participate in the relaxation mechanisms.⁴⁷ The transformation of the amorphous zones into crystalline zones makes the C–Cl polar bonds associated with these zones inactive at the same time. Consequently, it is expected that fewer dipoles compensate for surface-deposited charges for samples with the highest crystallinity. A lower SPD should thus occur but the reverse is observed. One can thus conclude that β -relaxation cannot explain the observed SPD.

A last lead must be studied: the distribution and dynamic of traps close to the surface are certainly strongly modified by the annealing process. In order to evaluate this assumption, thermally stimulated surface potential decay (TSSPD)⁴⁸ is carried out on electret parylene C materials. To this end, electrets were analysed after 12 days by charging to get rid of excess electrical charge carriers. Then, a temperature slope was applied to the samples, which was 2 K/min ranging from 20°C to 150°C. The results are presented in Figure 11.

As-deposited sample (A) shows an initial drop-off at around 62°C, which can be attributed to the impact of the β -relaxation and the moving of the C–Cl dipoles leading to the easing of charge carrier decay. The sample retains a substantial charge until a second drop-off occurring at a temperature of around 110°C, not far above the glass transition temperature T_g of parylene C. A correlation between the stability of SPD and the crystalline nature of the material is not so obvious. However, one can see that the higher the average area of surface crystallites (see Figure 9), the higher the temperature stability in SPD that is obtained (Figure 11). This correlation is not surprising as ions are deposited on the surface of the material after corona discharges.

To be able to identify the different traps at the origin of the TSSPD behavior, differentiating the curves is recommended with respect to the temperature according to eq. (4) to obtain the released current as a function of the temperature.⁴⁸

$$I(T) = \frac{\epsilon_0 \epsilon_r}{d} \Delta \frac{dV(T)}{dT} \quad (4)$$

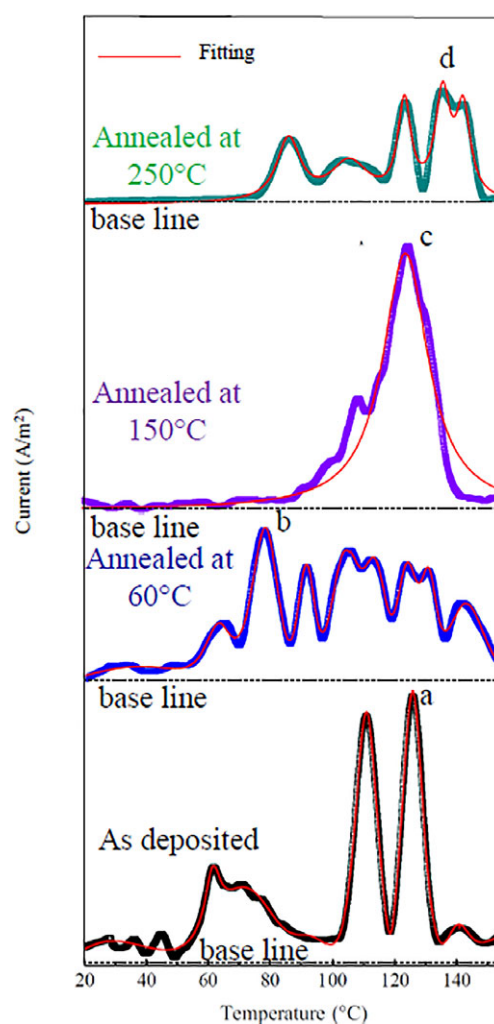


Figure 12. Fitting of the obtained released current resulting from the thermally stimulated surface potential decay for the different samples (the letters correspond to the maximum amplitude of the current, $a = 1.41 \times 10^{-6}$ A/m² for the as-deposited parylene C, $b = 7.77 \times 10^{-7}$ A/m² for the annealed sample at 60°C, $c = 1.52 \times 10^{-6}$ A/m² for the annealed sample at 150°C and $d = 6.45 \times 10^{-7}$ A/m² for the annealed sample at 250°C). [Color figure can be viewed at wileyonlinelibrary.com]

where ϵ_r is the relative permittivity of the parylene C, Δ is the heat rate temperature, d is the electret thickness and $V(T)$ is the measured SP as a function of the temperature.

Figure 12 demonstrates the thermally stimulated current for all samples at a heating rate of 2 K/min, along with the fits describing the released current up to 150°C. The fits are based on differential eq. (5), which describes the general second-order reaction kinetics.⁴⁸

$$I(T) = n_0 \exp \left(-\frac{\nu}{\Delta} \int_0^T \exp \left(-\frac{E_a}{k_b T'} \right) dT' \right) \nu \exp \left(-\frac{E_a}{k_b T} \right) \quad (5)$$

This equation assumes that no re-trapping phenomena occur in the material during the thermally stimulated discharge. As ν is the attempt-to-escape frequency, it should be fixed at

$\nu = 10^{13} \text{ s}^{-1}$ as ν is normally in the interval between 10^{13} and 10^{14} s^{-1} to get a meaningful fit. n_0 is the initial number of retained electrical charge carriers in the material before proceeding to the TSSPD. The activation energies E_a are evaluated from the temperature position of the current peaks using a numerical approach.

The existence of different peaks is representative of various traps with specific depths. In this fitting, peaks below 50°C are not considered as, in accordance with the TSSPD presented in Figure 11, noise in the measured signal and sensitivity of the electrostatic probe induces little derivative of the signal. The magnitude of the peaks depends on the retained trapped charges. Activation energies extracted from eq. (5) are summarized in Figure 13. Clearly, annealing strongly influences the distribution of traps inside the material and we can separate this impact in different ways. In terms of magnitude, virgin sample A and the sample that presents a higher surface crystallite size and roughness (sample C) retained the maximum number of trapped charges after corona discharge. Apart from sample C, other samples show discrete peaks, thereby enabling the identification of many traps with allocated activation energies.

Just two peaks are identified for sample C. The broad temperature range where these peaks appear is probably representative of a continuum of traps with close activation energies. A higher number of spherulites for this material could induce important strain with reorganization of trap density. Such an effect could also explain the identification of specific traps following the parylene C annealing.

Charges located at the amorphous-crystalline interfaces could also participate in this current [Maxwell–Wagner–Sillars (MWS) relaxation]. However, if such a mechanism takes place, it cannot be considered to be predominant here as the activation energy for MWS was identified as being more than 1.5 eV ⁴⁹ whereas in our case, values lower than 1.3 eV were obtained.

Moreover, no obvious correlation is identified between the dynamic of trapped charges created by corona and the dynamic of polymeric chains inside the polymer. Indeed, Figure 14 reports the dielectric response (ϵ' and ϵ'') of parylene C at 120°C (temperature in the range of the strong decay observed in Figure 11). For example, samples A and B are close in terms of their dielectric response but they are extremely different as regards their TSSPD.

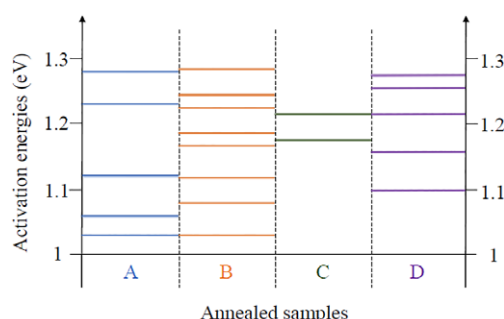


Figure 13. Extracted activation energies for the electret parylene C. [Color figure can be viewed at wileyonlinelibrary.com]

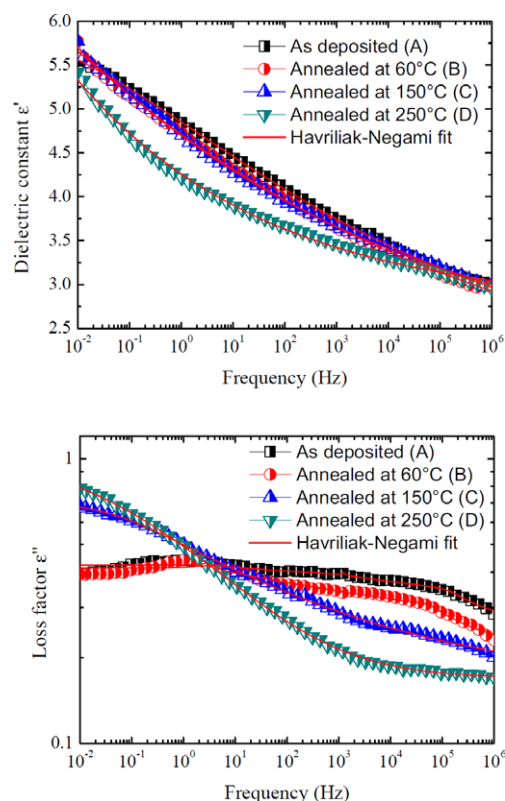


Figure 14. Dielectric constant ϵ' and loss factor ϵ'' versus frequency at 120°C for parylene C before and after annealing. [Color figure can be viewed at wileyonlinelibrary.com]

Dielectric analysis just confirms that annealing strongly modified the crystallinity, thereby inducing more or less mobility of molecular chains as shown in Figure 15 for m and n parameters for each sample. Both these parameters are extracted from the exponents of the Havriliak–Negami relation expressed by eq. (6).⁵⁰

$$\epsilon^*(\omega) = \epsilon_\infty + \frac{\epsilon_s - \epsilon_\infty}{[1 + (i\omega\tau)^a]^b} \quad (6)$$

and $m = a$ and $n = a.b$.

The parameters a and b represent, respectively, the symmetric and asymmetric nature of the relaxation peaks. ϵ_s and ϵ_∞ are the

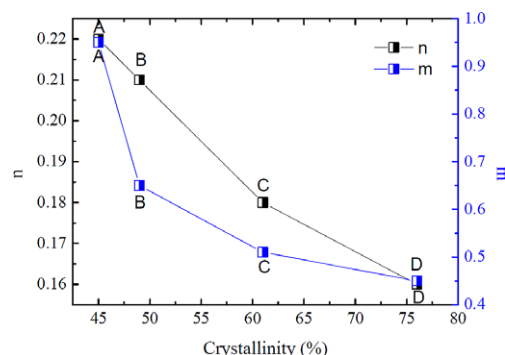


Figure 15. HN parameters as a function of the crystallinity at 120°C . [Color figure can be viewed at wileyonlinelibrary.com]

static dielectric constant and the high-frequency dielectric constant, respectively. ω is the angular frequency. τ is the relaxation time characteristic of the relaxation process.

The parameter n represents the dynamics of the short chains participating in the α -relaxation. The stress effect imposed by the crystallinity in the movement of these chains will therefore be low, which explains why n varies from 0.22 for $\chi_c = 45\%$ to 0.16 for $\chi_c = 75\%$. At the same time, the parameter m , which represents the dynamic of the long chains, drops from 0.95 to 0.45 and is coherent with a higher freedom of movement for chains at lower crystallinity.

This research work demonstrates extended potentialities for electret-based parylene C in terms of charges retained and thermal stability. This was possible after a simple controlled annealing of material.

CONCLUSION

We reported an improved electret-based parylene C polymer. Parylene C with a thickness of 4.5 μm , manufactured by CVD, was then annealed at various temperatures and changes in crystallinity, chemical structure and morphology were discussed. SPD measurements after corona discharge allowed evaluating performance of the parylene C as electret. Annealing at around 150°C for 6 h allows stable corona-deposited surface charges for energy harvesting electrostatic generators to be obtained. Furthermore, thermal stability of this electret is demonstrated up to 100°C. Supporting by XRD and dielectric properties, we concluded that spherulite area and trap density arrangement due to crystalline changes induced by annealing strongly influence this improvement. Finally, this work opens up new possibilities for developing electrets with complex shapes thanks to these new potentialities of compliant parylene C polymers.

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